higher frequencies, so that energy storage requirement is decreased, allowing smaller power electronics to be fitted. This makes it possible to reduce the size of the couplers and increase the power transfer density and displacement current.

The operating frequencies are kept below 100kHz in the inductive systems to minimise losses in the ferrite cores that are used for magnetic flux guidance and shielding. This results in larger coils and lower power transfer densities, making them bulky.

Ferrites are not required in the capacitive systems as the directed nature of electric fields greatly reduces the need for electromagnetic field guidance and shielding. The system can be operated at frequencies higher than 100kHz, allowing the size to be reduced. At higher operating frequencies the losses increase and achieving high efficiency becomes a challenge.

Ensuring human safety

Humans and animals may be present in the vehicle cabin or immediate vicinity of the perimeter of the chassis. They need to be protected against the electromagnetic radiations. For this purpose, the fringing electromagnetic fields of the system are required to be maintained within the safe levels laid down by the International Commission on Non-Ionizing Radiation Protection (ICNIRP 1998).

In capacitive systems the requirement is met by designing circuit stages to optimise voltage and current gain and by using reactive compensation. Other approaches used by the designers include employing field-focusing phased-array multiple phase-shifted capacitive modules and minimising the fringe radiations by beam forming.

Optimising power transfer

The systems are designed to operate close to the resonant frequency of the coupler reactance and the compensating network to optimise power transfer. The coupler reactance depends on the vehicle's road clearance and varies as the vehicle moves on the road across the charger, causing drift between the operating and resonant frequency. The drift causes a reduction in power transfer and system efficiency.

BMW static wireless charging pad for its electric cars (Credit: www.theverge.com)

To maintain the resonant frequency unchanged as the transmitter and receiver move relative to each other, the preferred technique is to use high-frequency rectifiers and inverter compensation architectures. They compensate for coupling variations to maintain the operating frequency at a fixed level. This ensures that the output power of the system is maintained at a fixed level across wide variations in coupling and is applicable to both capacitive and inductive systems.

For systems operating at frequencies below 100kHz, the frequency band is not a restriction and the systems can be designed such that the operating frequency tracks the resonant frequency. In high-frequency systems this is not feasible as the operating frequency has to stay within the designated industrial, scientific, and medical (ISM) bands.

Wireless charging systems available

There are basically two types of charging systems—static and dynamic.

Static charging systems. Several static charging systems are now available. BMW is providing a wireless charging pad for its electric car 530e iPerformance. The pad can be connected to the power outlet in the garage to charge the car wirelessly. About 3.5 hours are normally required to fully charge the 9.5kWh battery of the car.

Wireless chargers for Tesla Model S, BMW i3, Nissan Leaf, and first-generation Chevrolet Volt are readily available in the market. The cost of these chargers varies from $1,260 to $3,000. These can enable 40km to 80km of per hour of charging, depending on the system used.

Another wireless charging system called Plugless System available in the market includes a charger that rests on the floor, and a power receiver that is attached to the car. The receiver can be removed when one wants to keep the charger while disposing of the car.

Another wireless charging system available in the market is the Halo system, which is becoming increasingly popular with the electric emergency cars deployed for Formula E races. The system keeps the batteries of the BMW i3 medical cars and i8 safety cars topped up during each race day. Since the cars are not required to be plugged in, no valuable seconds are wasted when the cars are needed out to attend an emergency on circuit. This time saving is crucial for electric ambulances and other emergency vehicles.

IN2POWER, a Belgian innovator, has recently brought out 1.1kW to 16kW 'plug & play' wireless charging systems (combinable up to 48kW) for autonomous guided vehicles, drones, medical and naval applications to charge batteries with 30A to 750A current with 95% efficiency. Wiferion, a